Mengqing Liu

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Hamilton, ON, Canada ✻ Authorized to work in Canada

PROFESSIONAL SUMMARY

Applied AI and machine learning professional with a Master's degree in Computer Technology, and hands-on experience developing end-to-end AI workflows that integrate task-specific deep learning models, Large Language Model (LLM) agents, and human review. Skilled in Python-based data processing, model inference, quality assurance, and application integration, translating model outputs into structured findings, review tasks, and actionable feedback. Experienced in rubric-based Vision-Language Model (VLM) evaluation, dataset quality review, failure analysis, and incorporating expert feedback into iterative system improvement. Published computer vision researcher. Native Mandarin Chinese speaker with English working proficiency.

CORE COMPETENCIES

- AI Training Data Labeling & Quality Review
- Data Cleaning, Validation & Transformation
- VLM & Instruction-Following Evaluation
- Model Output Review & Error Analysis
- Visual Data Annotation & Ground Truth
- Attention to Detail & Independent Work
- Python, Pandas, NumPy
- APIs, Web Scraping, JSON, CSV
- MySQL, SQLite, Data Organization
- PyTorch, OpenCV, scikit-learn
- Technical Documentation & Reporting
- Stakeholder Communication & Collaboration

PROFESSIONAL EXPERIENCE

TalentAI *Remote*
AI Evaluation Contributor (Part-time) *May 2026 – October 2026*

- Evaluated image-grounded VLM tasks and AI-generated responses against detailed rubrics, assessing instruction following, visual reasoning, factual consistency, and overall response quality.
- Reviewed training and evaluation data for ambiguity, missing information, labeling consistency, and scenario relevance; identified edge cases and documented quality issues clearly.
- Applied scenario labels, quality scores, and structured written feedback to support dataset organization, model benchmarking, and downstream model improvement.
- Worked independently on English- and Chinese-language evaluation tasks while maintaining careful and consistent judgments across repeated reviews.

Nantong Institute of Technology *Nantong, China*
Applied AI Engineer *Aug 2023 – Jul 2025*

- Developed LLM-powered assessment agents to evaluate student responses against instructor-defined rubrics and generate criterion-level scores, supporting evidence, and feedback drafts.
- Built a human-in-the-loop grading pipeline integrating submission processing, rubric-based scoring, and instructor review for assignments and exams.
- Implemented Python services and APIs for submission ingestion, rubric configuration, score review, and grade export, maintaining traceable assessment records.
- Added validation checks for missing responses, score limits, and inconsistent feedback, flagging ambiguous cases for instructor review.
- Evaluated grading agreement, error patterns, and review time against instructor assessments, using reviewed cases to refine prompts and evaluation workflows.

PROFESSIONAL EXPERIENCE – CONTINUED

Nanchang Hangkong University *Nanchang, China*
Graduate Research Assistant *Jun 2020 – Jun 2023*

- Built an end-to-end deep learning pipeline integrating 3D image preprocessing, segmentation model training and inference, quantitative evaluation, and manual quality review to support iterative model refinement.
- Prepared, cleaned, transformed, and analyzed 3D prostate ultrasound datasets using Python, NumPy, PyTorch, and medical image processing tools.
- Created and validated ground-truth labels, performed preprocessing and augmentation, and maintained organized experimental data for model training and evaluation.
- Assessed model predictions using quantitative metrics and visual inspection, identified difficult cases and recurring error patterns, and refined experiments based on the findings.
- Collaborated with researchers on machine learning experiments and technical reporting and contributed to peer-reviewed publications in medical image analysis.

FEV Powertrain Technology (Shanghai) Co., Ltd. *Nanjing, China*
Project Manager Intern *Oct 2022 – Jun 2023*

- Researched AI and deep learning applications in autonomous driving and intelligent vehicle scenarios, organizing technical findings and use-case information for project evaluation.
- Supported testing progress, documentation, task follow-up, and cross-team communication in an Agile project environment.

EDUCATION

Master of Computer Technology 2020 – 2023
Nanchang Hangkong University, China

Bachelor of Internet of Things 2016 – 2020
Nanjing University of Information Science and Technology, Binjiang College, China

SELECTED PUBLICATIONS

[**1**] M. Liu, X. Shao, L. Jiang, and K. Wu, "3D EAGAN: 3D edge-aware attention generative adversarial network for prostate segmentation in transrectal ultrasound images," *Quantitative Imaging in Medicine and Surgery*, 2024.

[**2**] M. Liu, K. Wu, and L. Jiang, "ADC-Net: Adaptive detail compensation network for prostate segmentation in 3D transrectal ultrasound images," *Medical Imaging 2023: Ultrasonic Imaging and Tomography (SPIE)*, 2023.